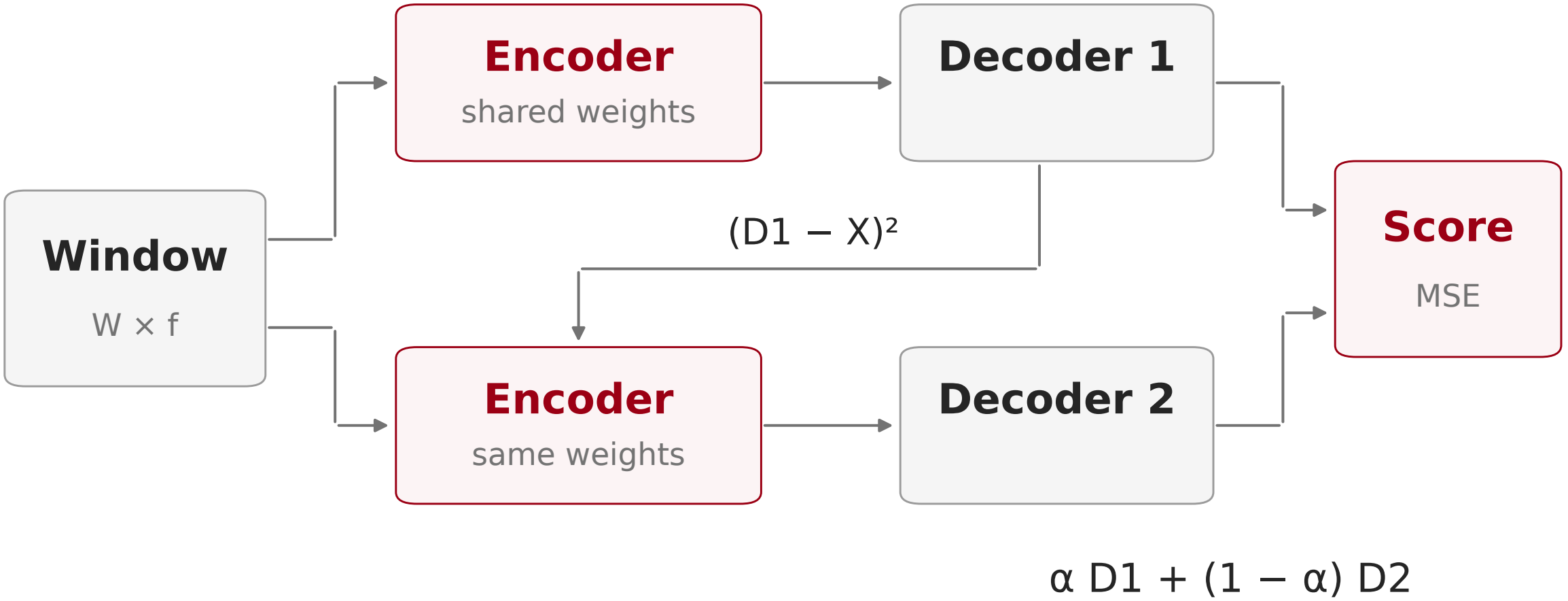
TranAD+


Encoder
shared weights
Decoder 1
Window
W × f
(D1 − X)²
Score
MSE
Encoder
same weights
Decoder 2
α D1 + (1 − α) D2

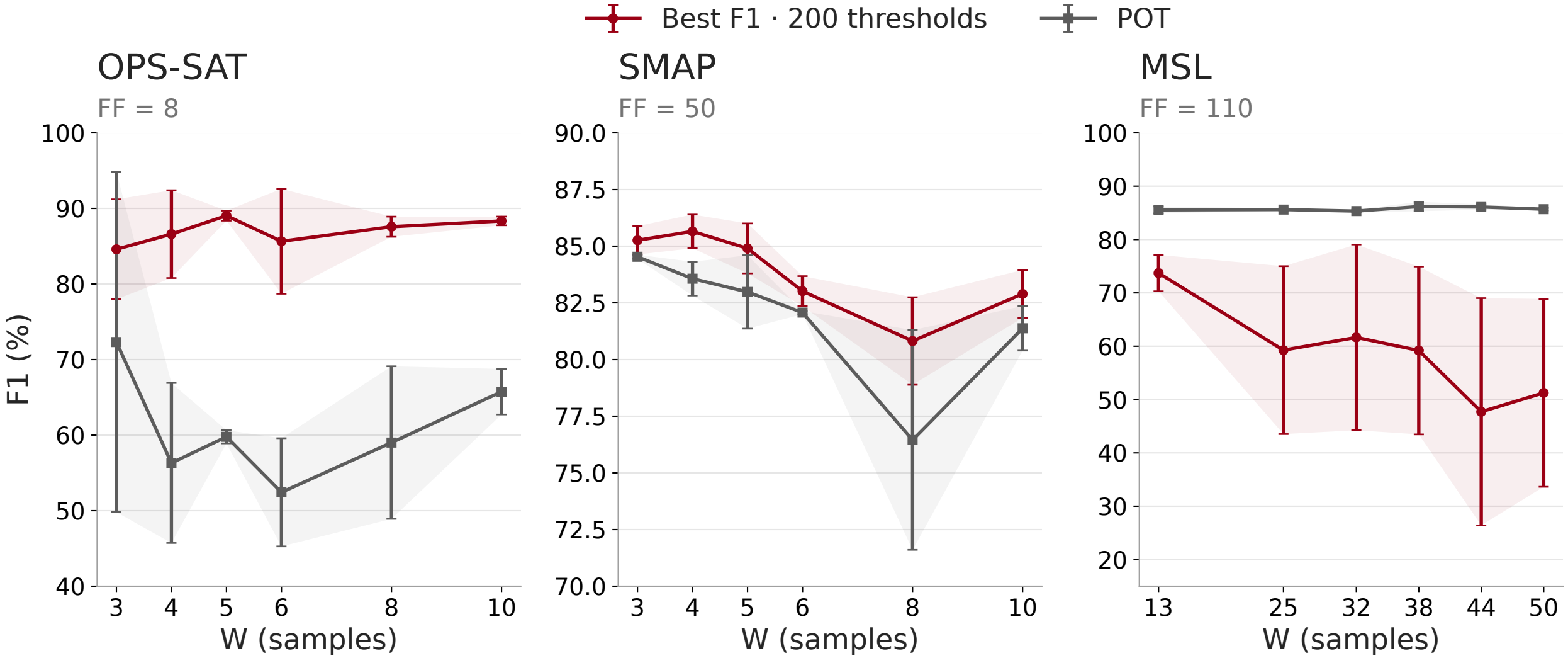
Benchmark reproduction · F1 (%)

	POT		Best F1 · 200 thresholds	
Dataset	Paper	Ours	Paper	Ours
DSN_1k	92.47	92.34	80.44	80.73
SMAP	84.24	84.22	84.91	84.70
MSL	89.01	90.17	81.96	79.03
SWaT	32.90	32.49	82.39	82.07
SMD	71.29	68.61	85.09	84.06
WADI	50.84	53.39*	27.88	27.88

* WADI POT: NaN masking applied.

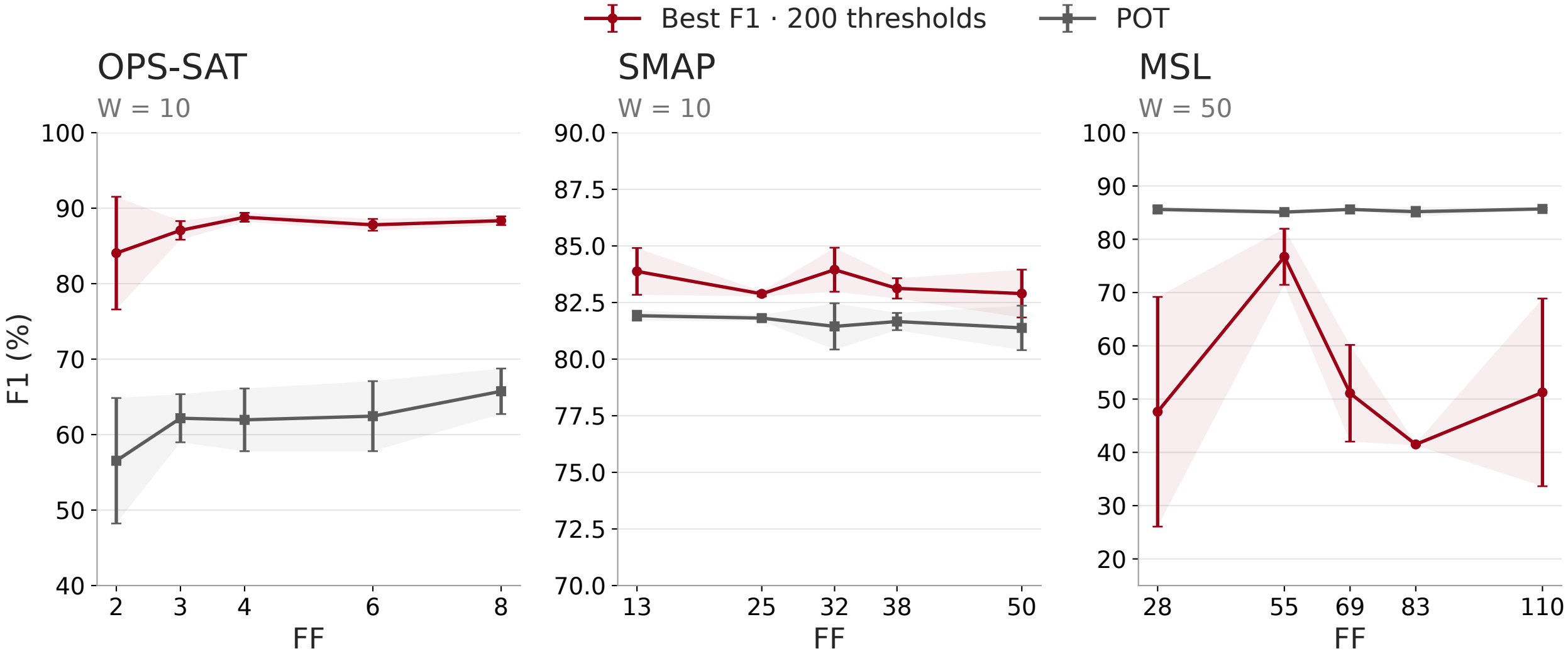
Point adjustment · best available individual runs, selected separately for each method

Window size



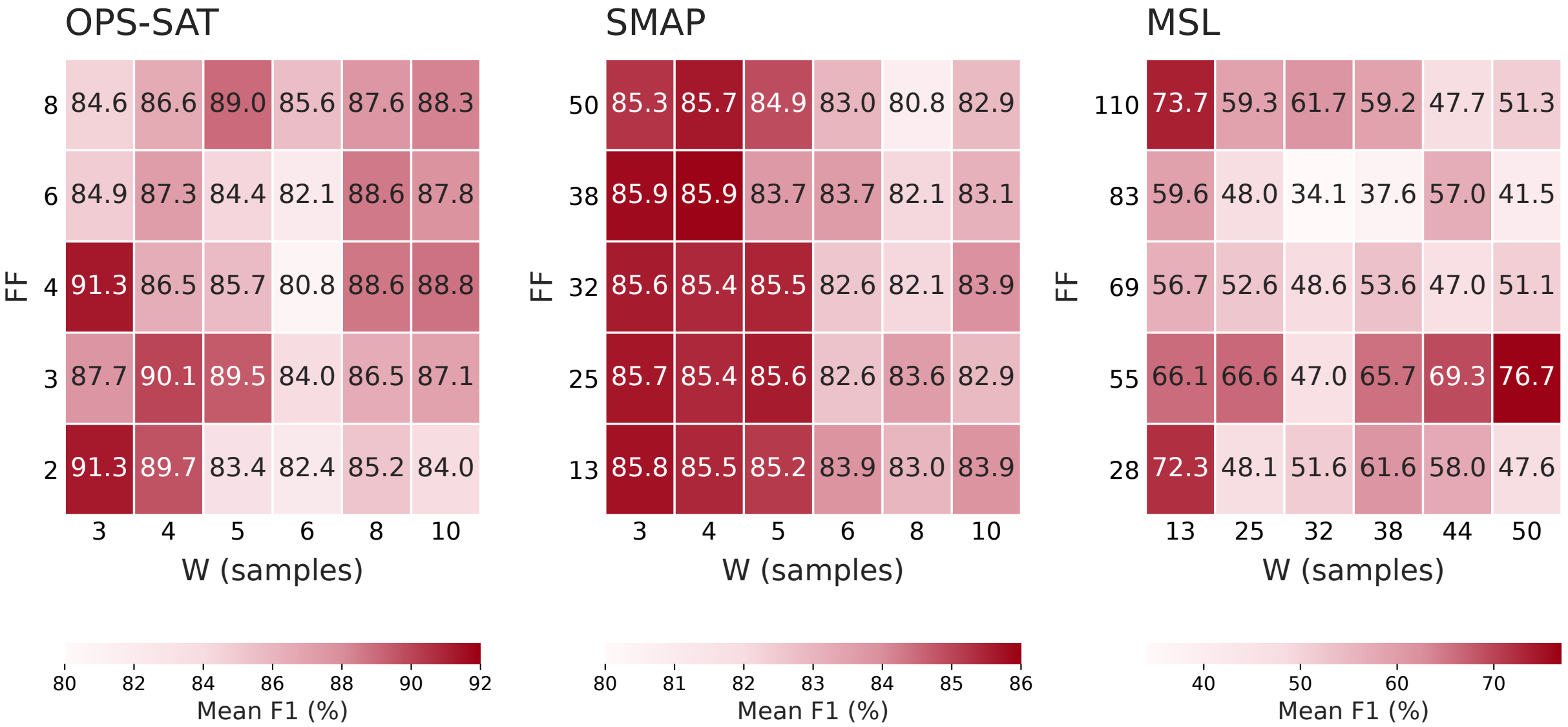
Mean ± sample SD · 3 seeds · 50 epochs · PA · dataset-specific vertical scales

Feed-forward dimension



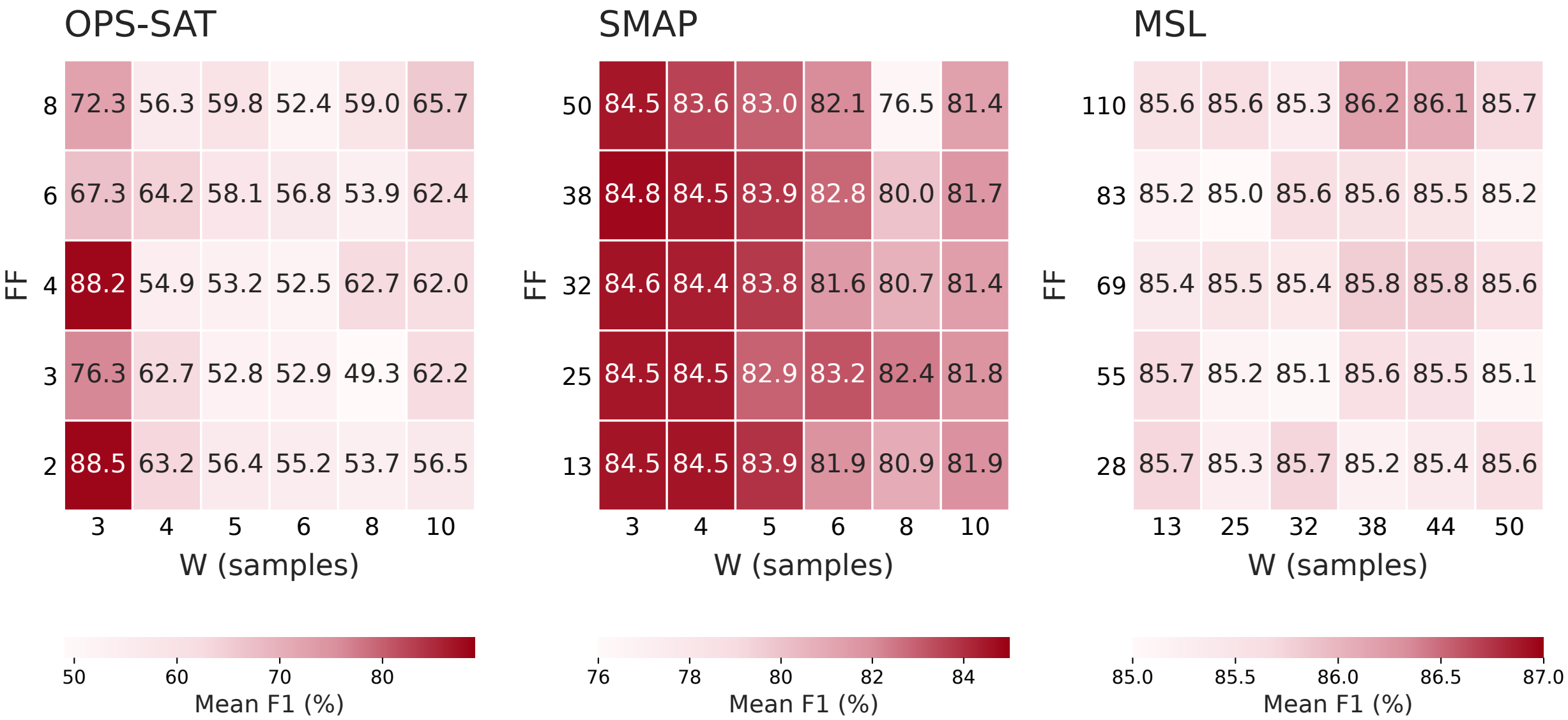
Mean ± sample SD · 3 seeds · 50 epochs · PA · dataset-specific vertical scales

Window × feed-forward dimension · Best F1 · 200 thresholds



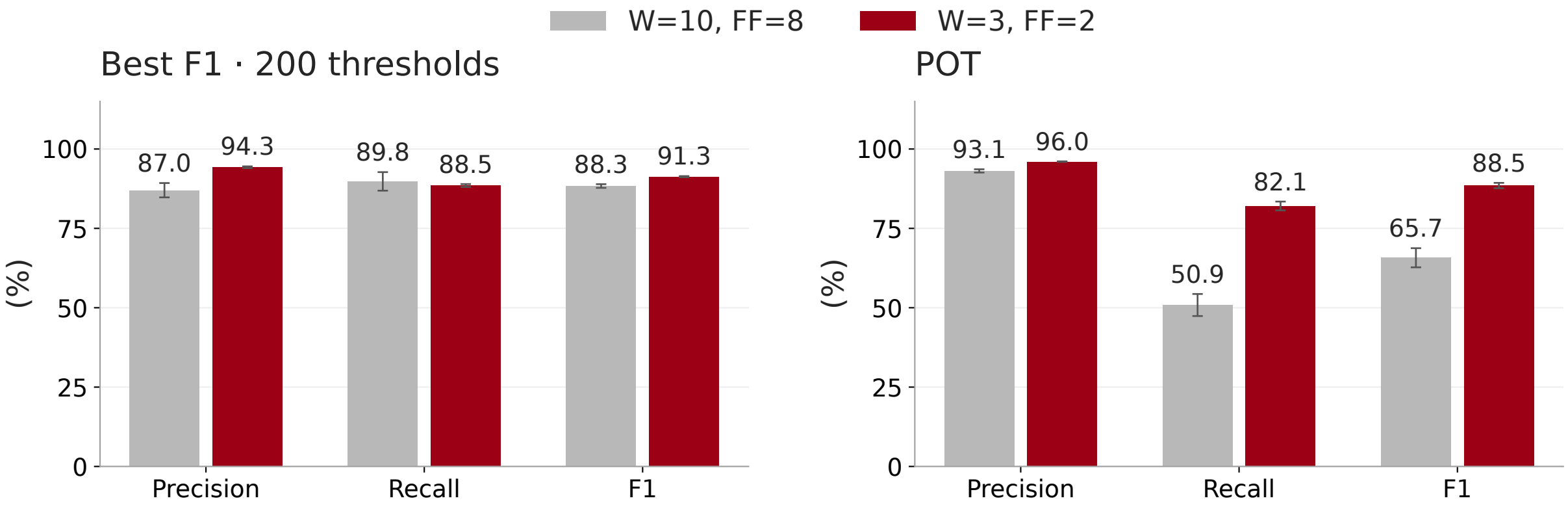
3 seeds per cell · 50 epochs · PA · separate colour scale for each dataset

Window × feed-forward dimension · POT



3 seeds per cell · 50 epochs · PA · separate colour scale for each dataset

OPS-SAT · what changes in the metrics?

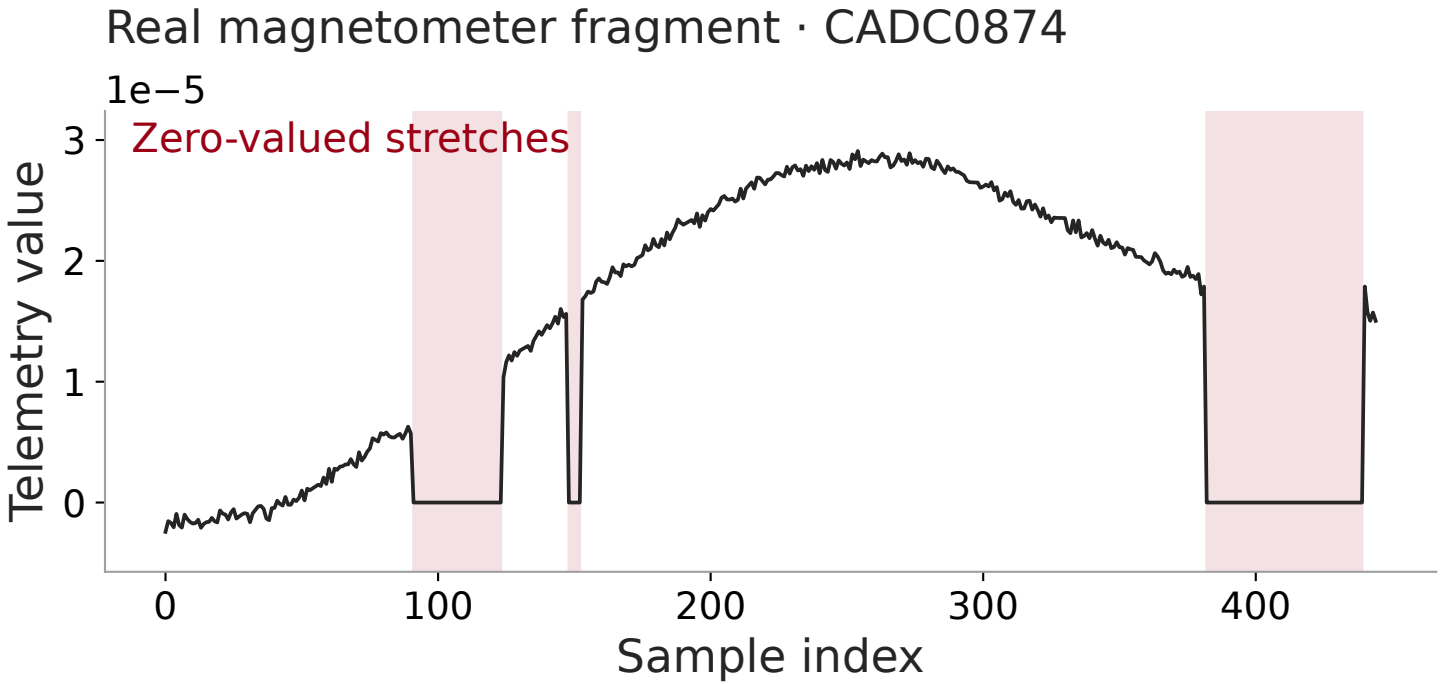


Reducing W and FF together improves both F1 scores.

BF200: higher precision, slightly lower recall. POT: recall also increases.

Mean ± sample SD · 3 seeds · 50 epochs · point adjustment

OPS-SAT · brief departures inside longer fragments



86%

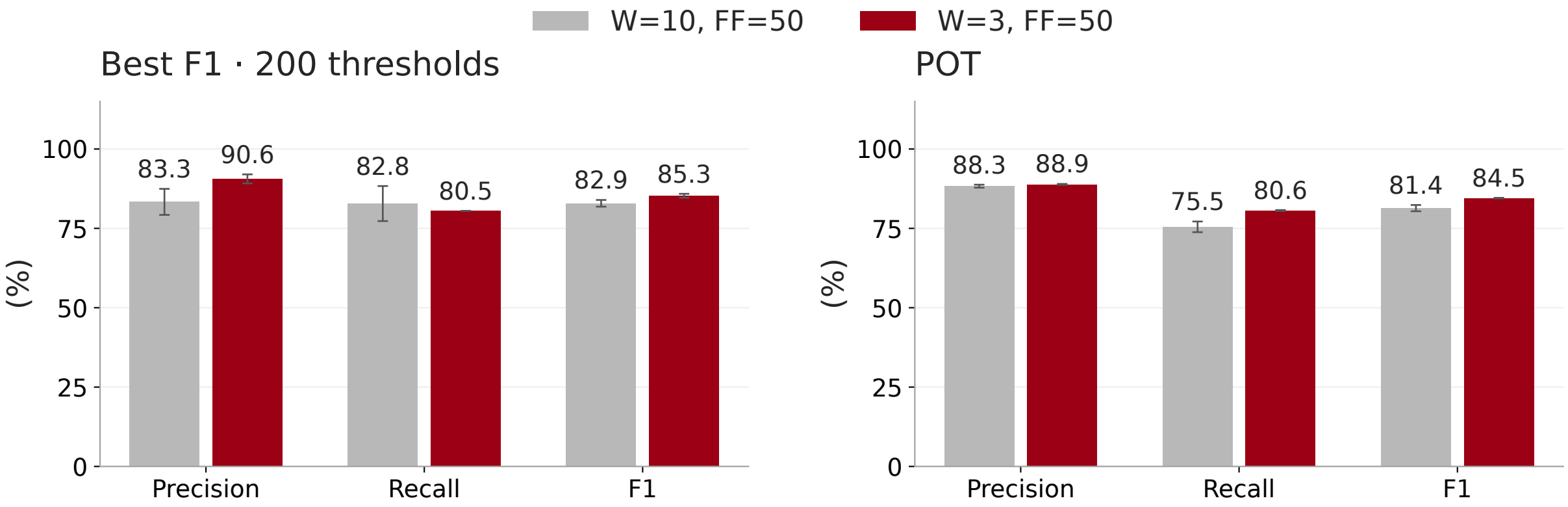
of positive-labelled samples come from magnetometers

3 magnetometer channels
6 photodiode channels

Brief departures can sit inside an otherwise ordinary-looking fragment.

A short window can change error peaks; extra history is not always helpful.

SMAP · what changes in the metrics?

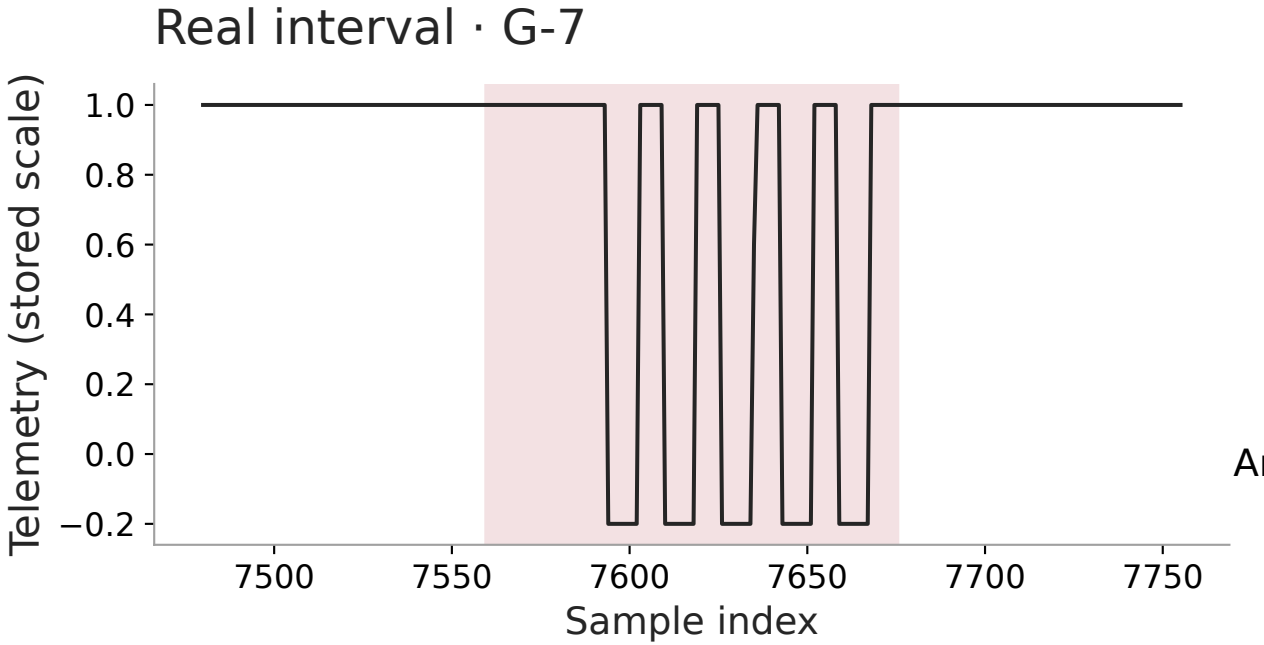


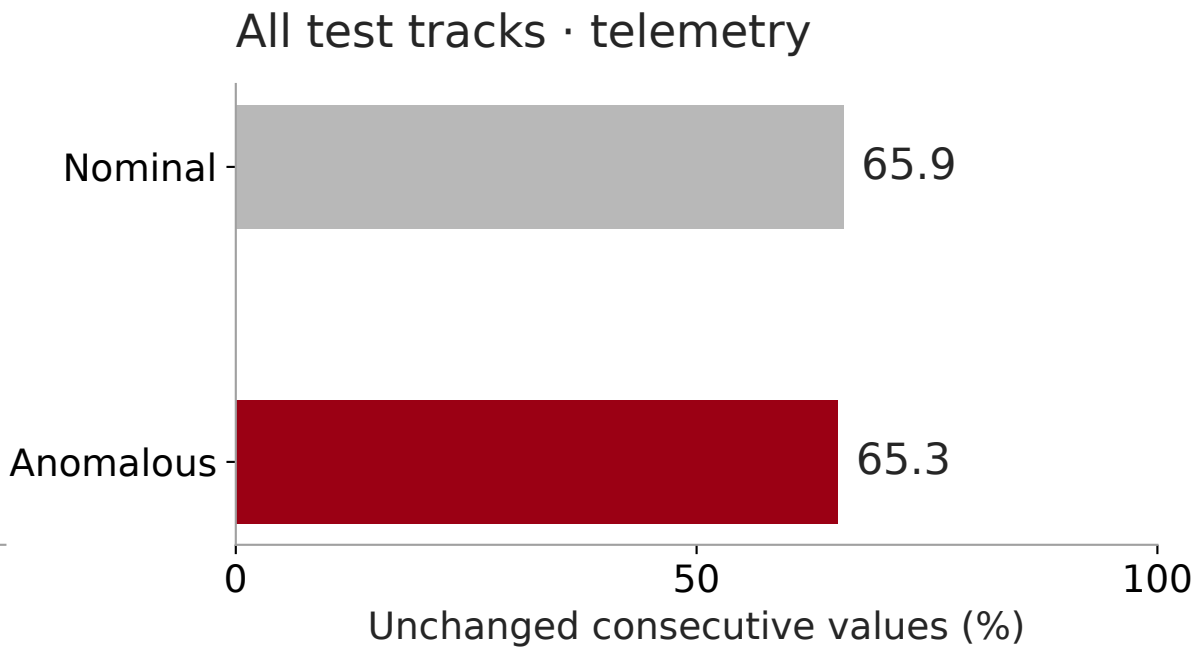
BF200 improves mainly through higher precision, with slightly lower recall.

POT improves mainly through higher recall.

Mean ± sample SD · 3 seeds · 50 epochs · point adjustment

SMAP · extra history can contain repeated information



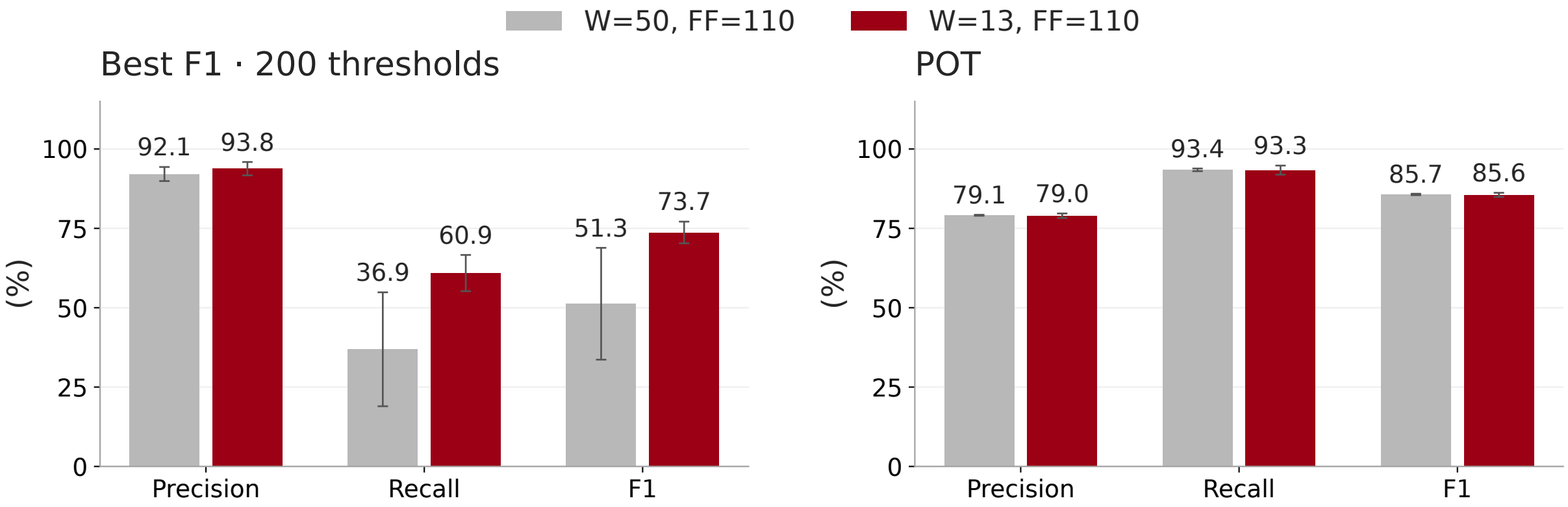


About two thirds of successive values are identical, in both classes.

Repeated values make redundancy plausible.

Real test data · shading = positive labels · telemetry coordinate only · example selection recorded in data/

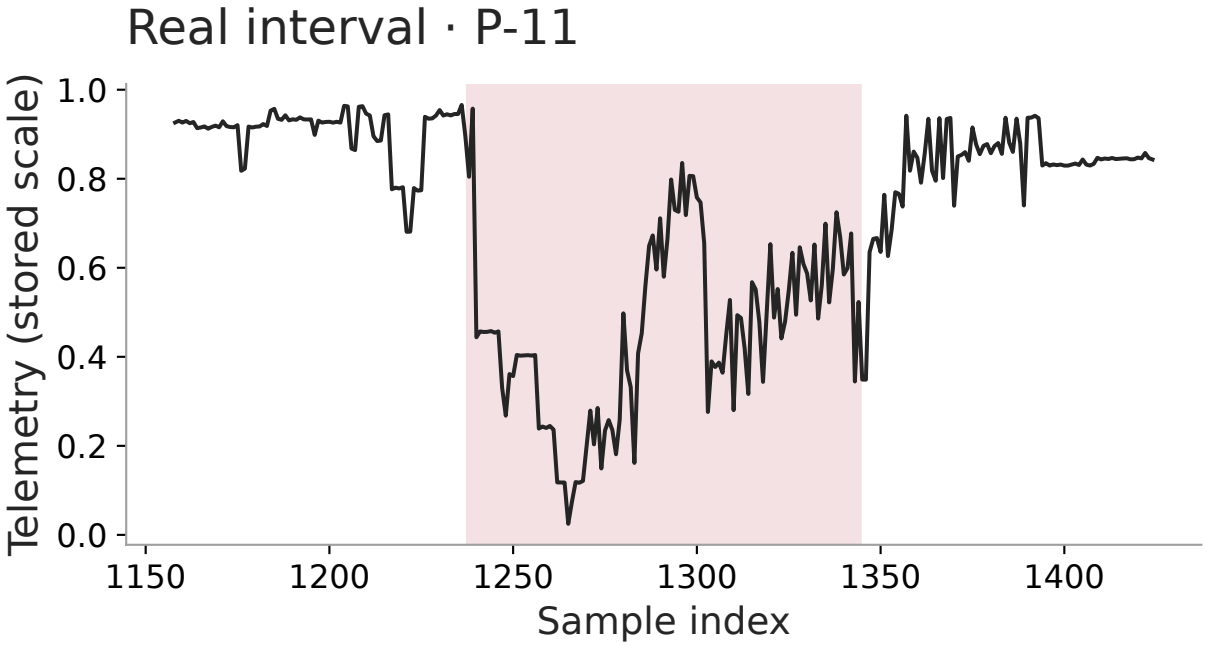
MSL · what changes in the metrics?

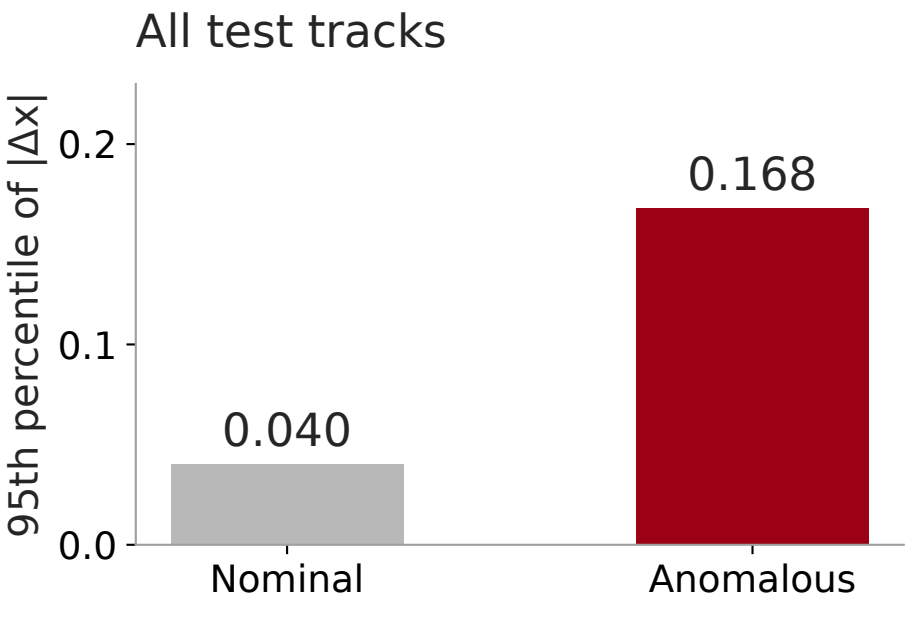


BF200 recall changes strongly; POT precision and recall stay almost unchanged.

Mean ± sample SD · 3 seeds · 50 epochs · point adjustment

MSL · anomalous intervals contain larger local changes

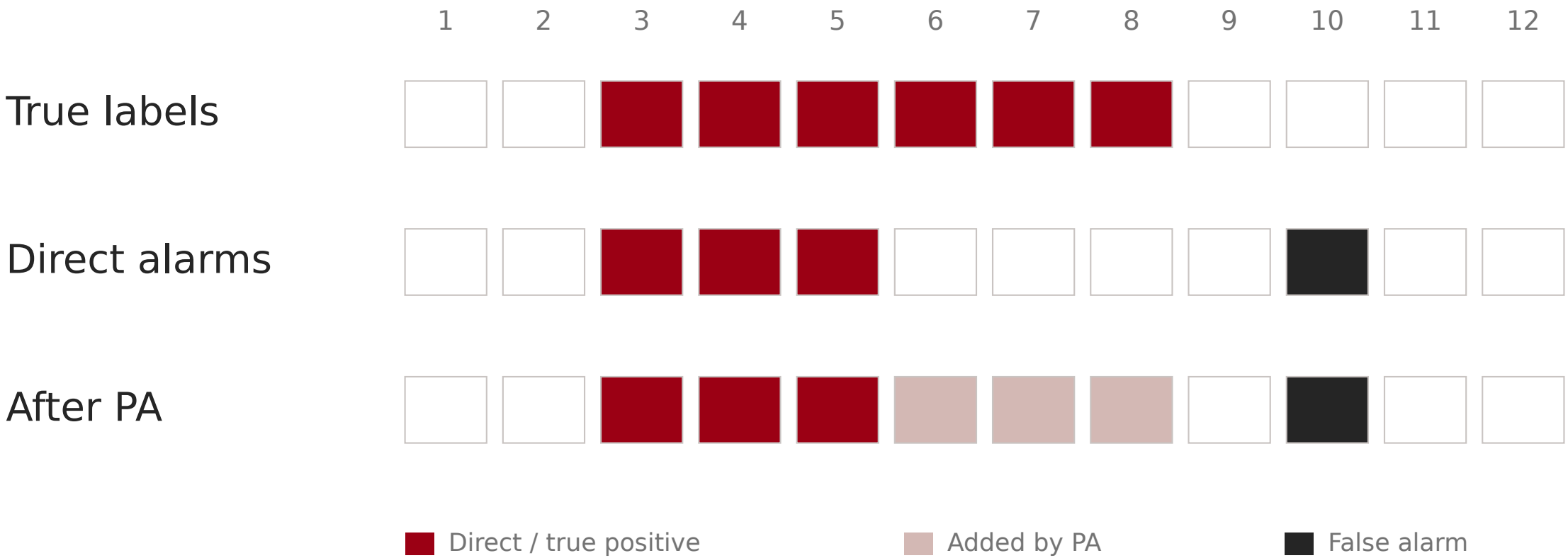




At the 95th percentile, local changes are about four times as large.

Real test data · shading = positive labels · right: |Δx| divided by nominal-training range

Point adjustment · what changes after an event is hit?

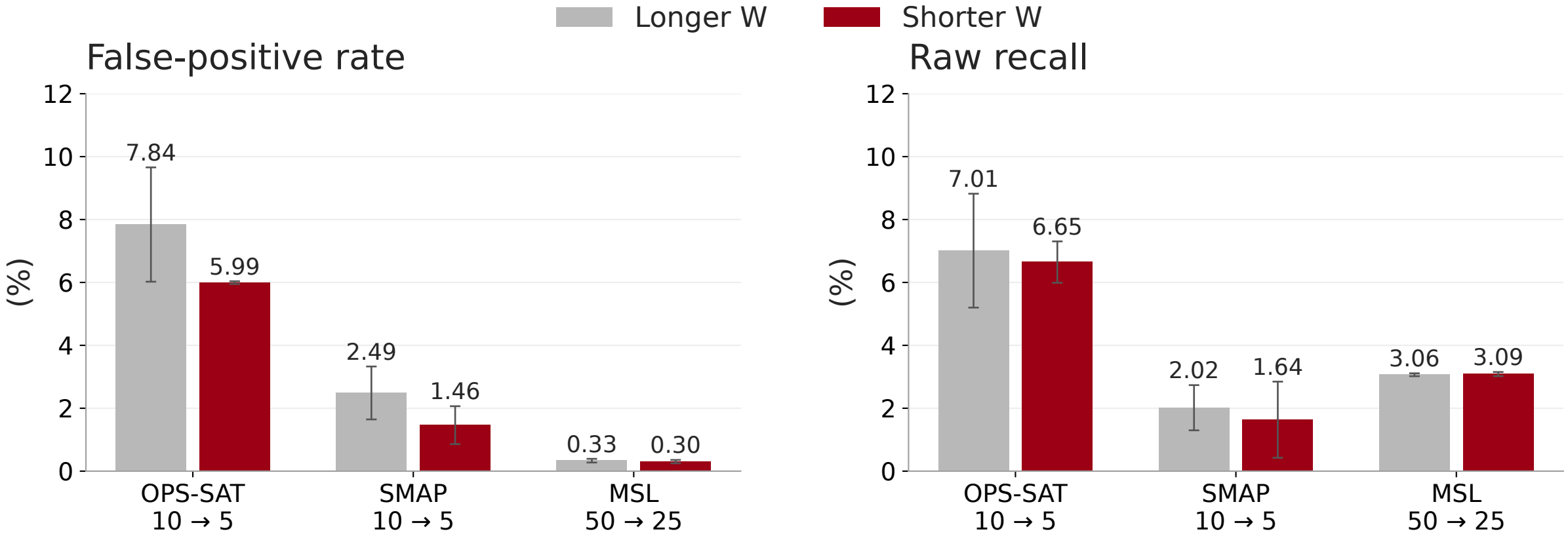


Before PA: TP = 3 FP = 1 FN = 3

After PA: TP = 6 FP = 1 FN = 0

Illustrative example · PA fills a detected positive interval; false alarms outside remain unchanged.

Fewer false alarms can also mean fewer raw detections

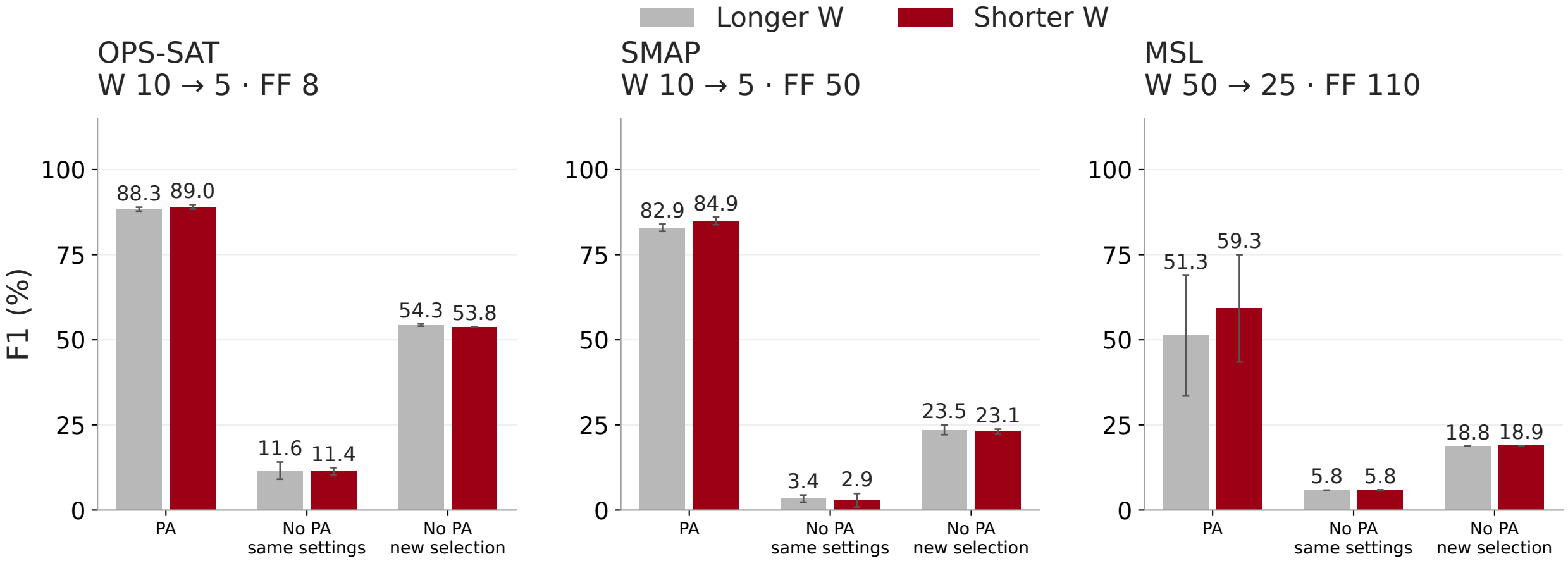


BF200: fewer false alarms, but also lower raw recall on OPS-SAT and SMAP.

PA leaves false-positive counts unchanged; it credits more positive samples.

Same PA-selected α and threshold for each model · 3 matched seeds · reference architecture · FF unchanged

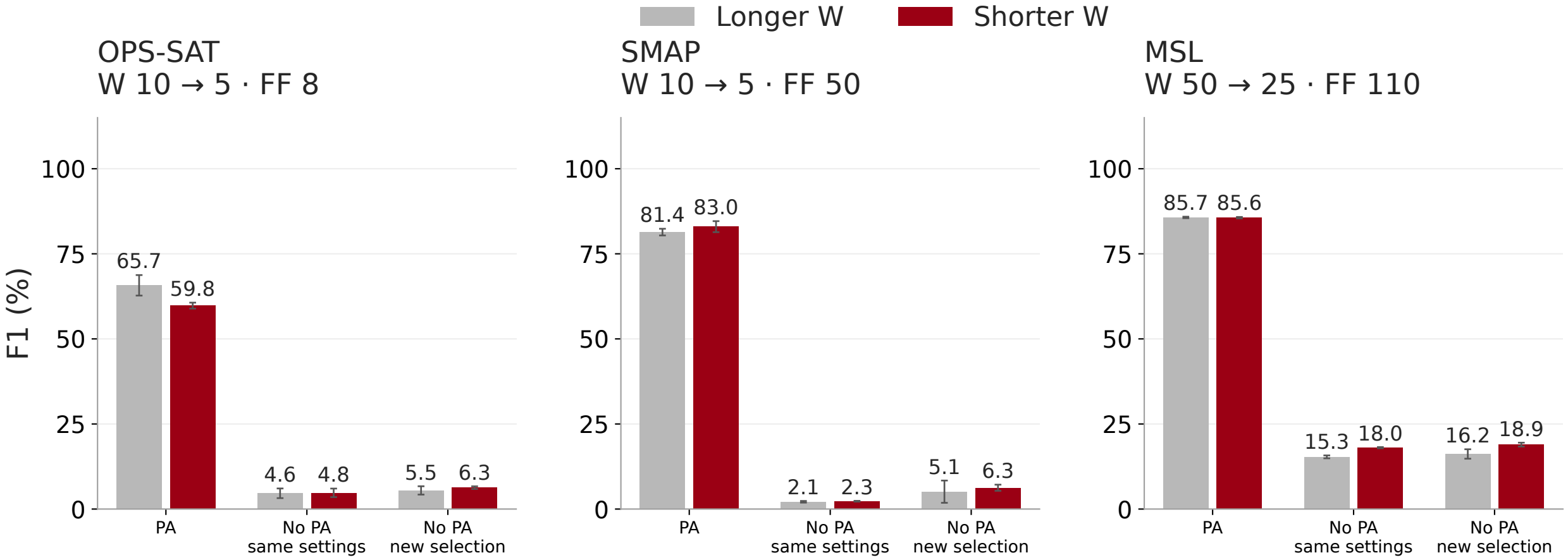
BF200 · with and without point adjustment



Same settings: remove PA, keeping the α and threshold selected with PA.

New selection: BF200 selects α and τ for raw F1 · same trained models · 3 seeds · 50 epochs

POT · with and without point adjustment



Shorter windows improve raw F1 here, but it remains much lower than PA F1.

Same settings: remove PA, keeping the α and threshold selected with PA.

New selection: POT selects α for raw F1 using its per-α SPOT threshold · same models · 3 seeds

What can we conclude about shorter windows?

Fewer false alarms can be a real benefit

But fewer alarms can also mean fewer directly detected anomalous samples.

PA changes what receives credit

A single crossing can credit a whole event; raw F1 counts individual samples.

Keep the two conclusions separate

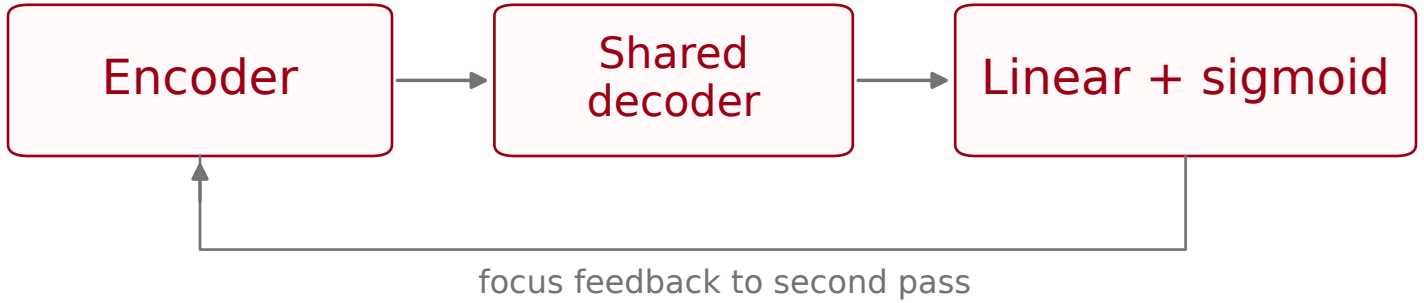
Better PA F1 does not necessarily mean better point-by-point detection.

Signal shape is a possible contributor; the matched controls show that method and evaluation also matter.

Architectural variants

One-decoder

2 passes · same encoder and decoder



One-pass

1 encoder–decoder pass



Encoder-only

1 encoder pass + reconstruction head



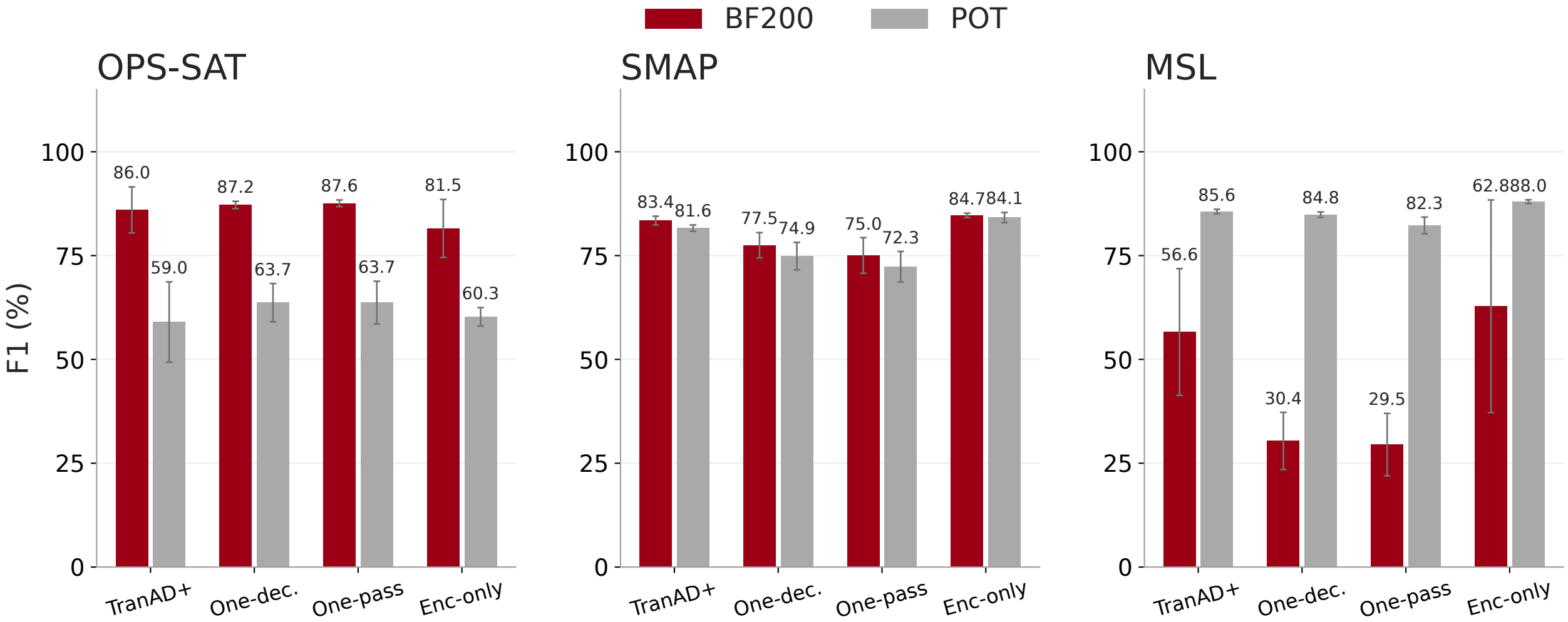
The reference TranAD+ has two passes and two decoders with separate parameters.

Architecture study · configurations

Dataset	Reference	Smaller W	Smaller FF	Both smaller
OPS-SAT	10 / 8	5 / 8	10 / 4	5 / 4
SMAP	10 / 50	5 / 50	10 / 25	5 / 25
MSL	50 / 110	25 / 110	50 / 55	25 / 55

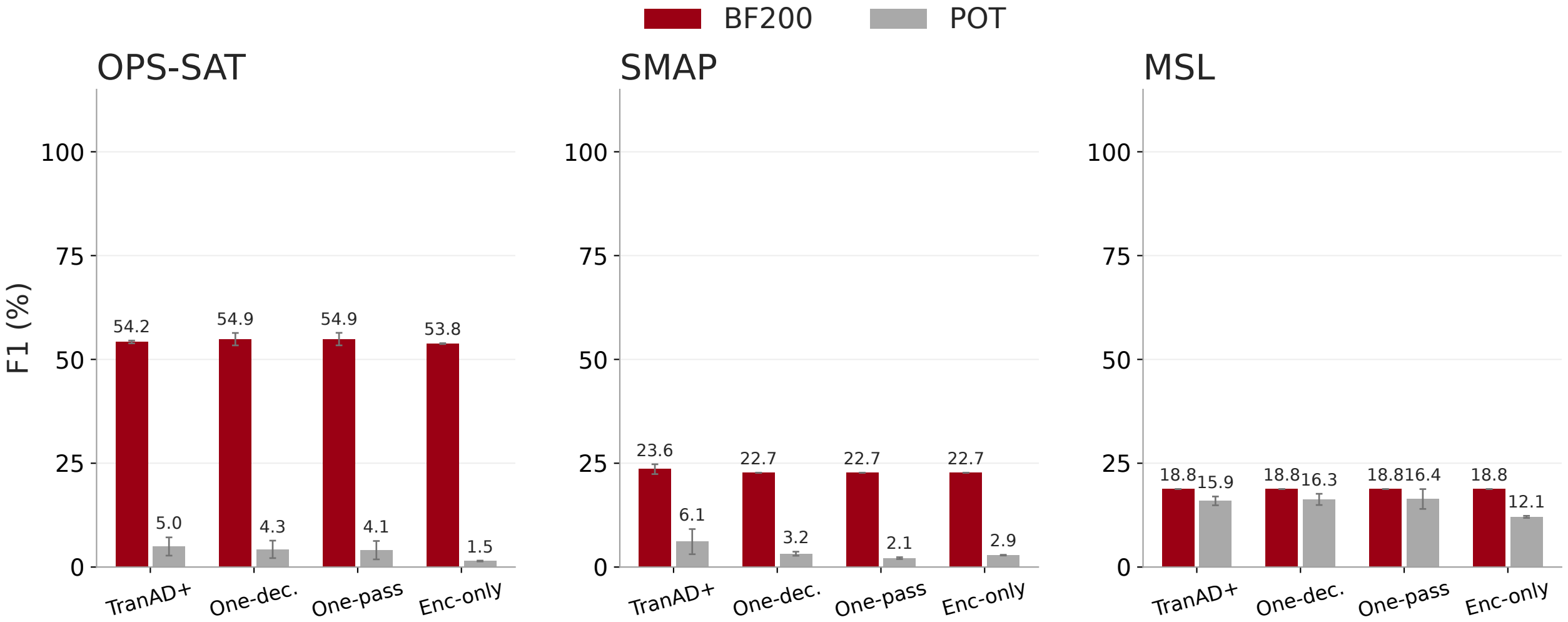
W / FF · 4 architectures × 4 settings × 5 seeds × 3 datasets = 240 trained models · 50 epochs

Architectures · F1 with PA



Reference W/FF · FP32 · mean ± sample SD over 5 seeds · selection targets PA F1

Architectures · F1 without PA



Reference W/FF · FP32 · mean ± sample SD over 5 seeds · selection targets raw F1

OPS-SAT · architecture results (%)

Model	Method	Precision	Recall	F1	AUC-PA
TranAD+	BF200	83.70	89.13	86.00	89.06
TranAD+	POT	92.20	43.92	59.01	70.91
One-decoder	BF200	86.78	87.66	87.19	89.92
One-decoder	POT	92.95	48.55	63.66	73.21
One-pass	BF200	87.43	87.80	87.60	90.22
One-pass	POT	92.95	48.58	63.66	73.23
Encoder-only	BF200	73.12	94.05	81.53	85.78
Encoder-only	POT	92.38	44.74	60.26	71.30

Reference W/FF · FP32 · PA · mean over 5 seeds · AUC-PA uses binary predictions after PA

SMAP · architecture results (%)

Model	Method	Precision	Recall	F1	AUC-PA
TranAD+	BF200	82.74	84.41	83.43	90.89
TranAD+	POT	88.18	75.96	81.61	87.23
One-decoder	BF200	88.91	70.59	77.50	84.50
One-decoder	POT	85.65	66.54	74.88	82.45
One-pass	BF200	86.84	66.67	75.03	82.55
One-pass	POT	87.11	61.94	72.30	80.30
Encoder-only	BF200	88.77	80.96	84.68	89.73
Encoder-only	POT	88.62	80.12	84.15	89.31

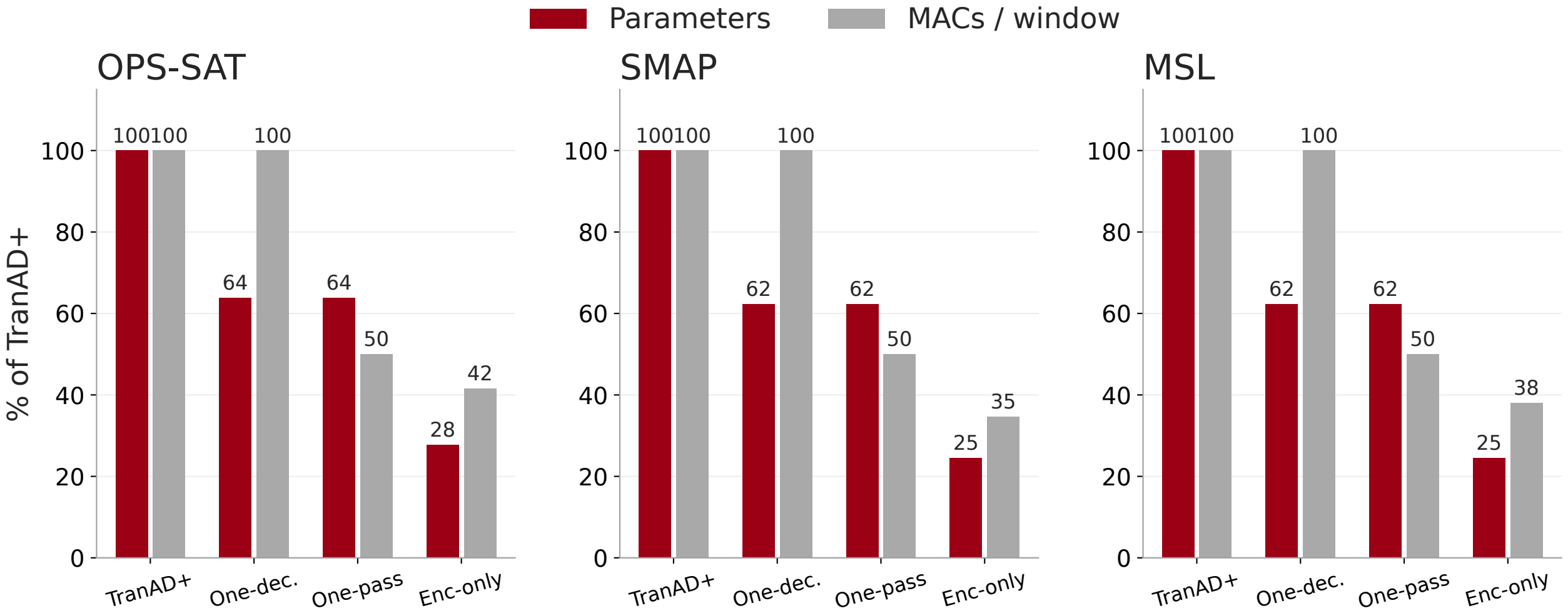
Reference W/FF · FP32 · PA · mean over 5 seeds · AUC-PA uses binary predictions after PA

MSL · architecture results (%)

Model	Method	Precision	Recall	F1	AUC-PA
TranAD+	BF200	92.32	42.11	56.60	70.87
TranAD+	POT	78.85	93.65	85.62	95.37
One-decoder	BF200	90.20	18.41	30.37	59.10
One-decoder	POT	77.57	93.64	84.85	95.26
One-pass	BF200	89.13	17.84	29.47	58.80
One-pass	POT	76.29	89.39	82.28	93.09
Encoder-only	BF200	90.78	52.29	62.79	75.89
Encoder-only	POT	83.90	92.51	87.99	95.23

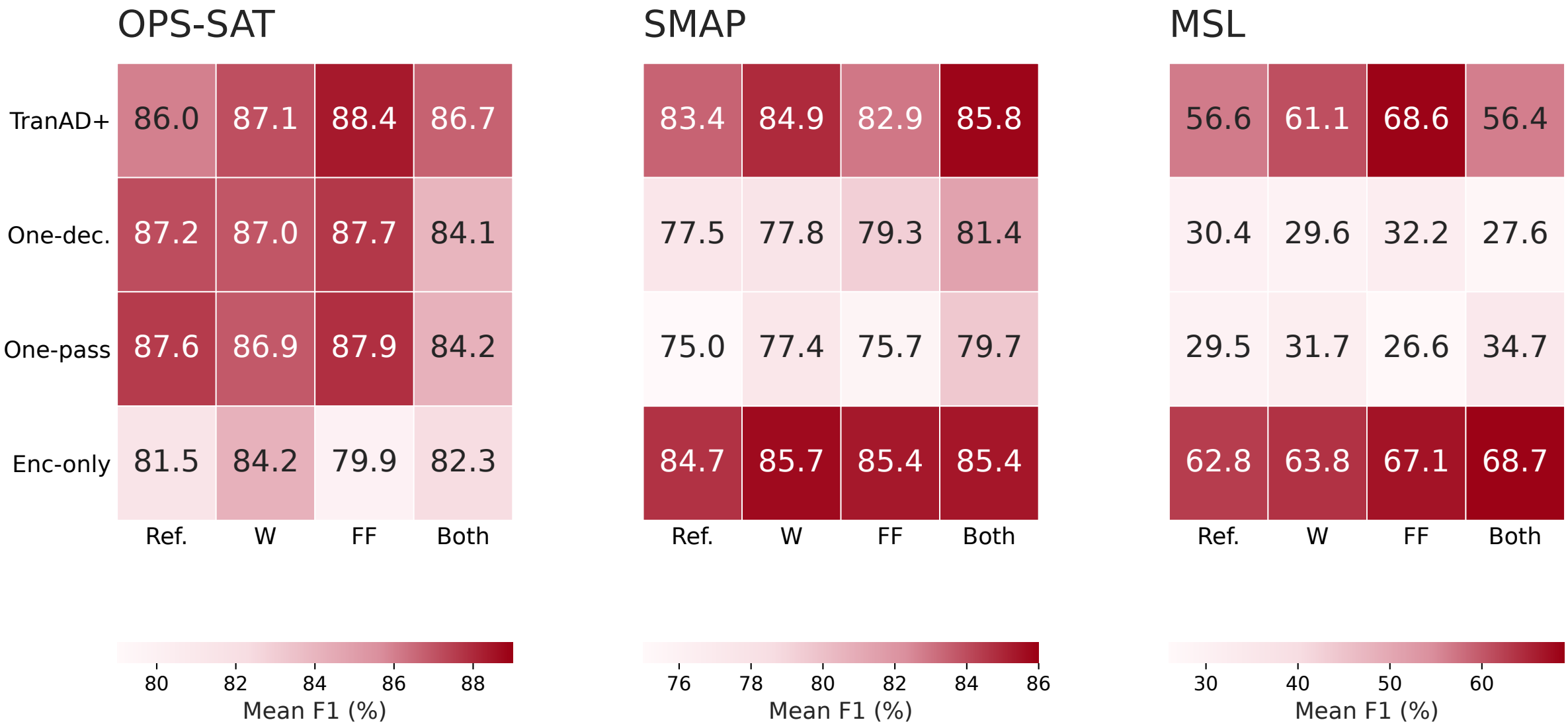
Reference W/FF · FP32 · PA · mean over 5 seeds · AUC-PA uses binary predictions after PA

Architectures · parameters and matrix operations



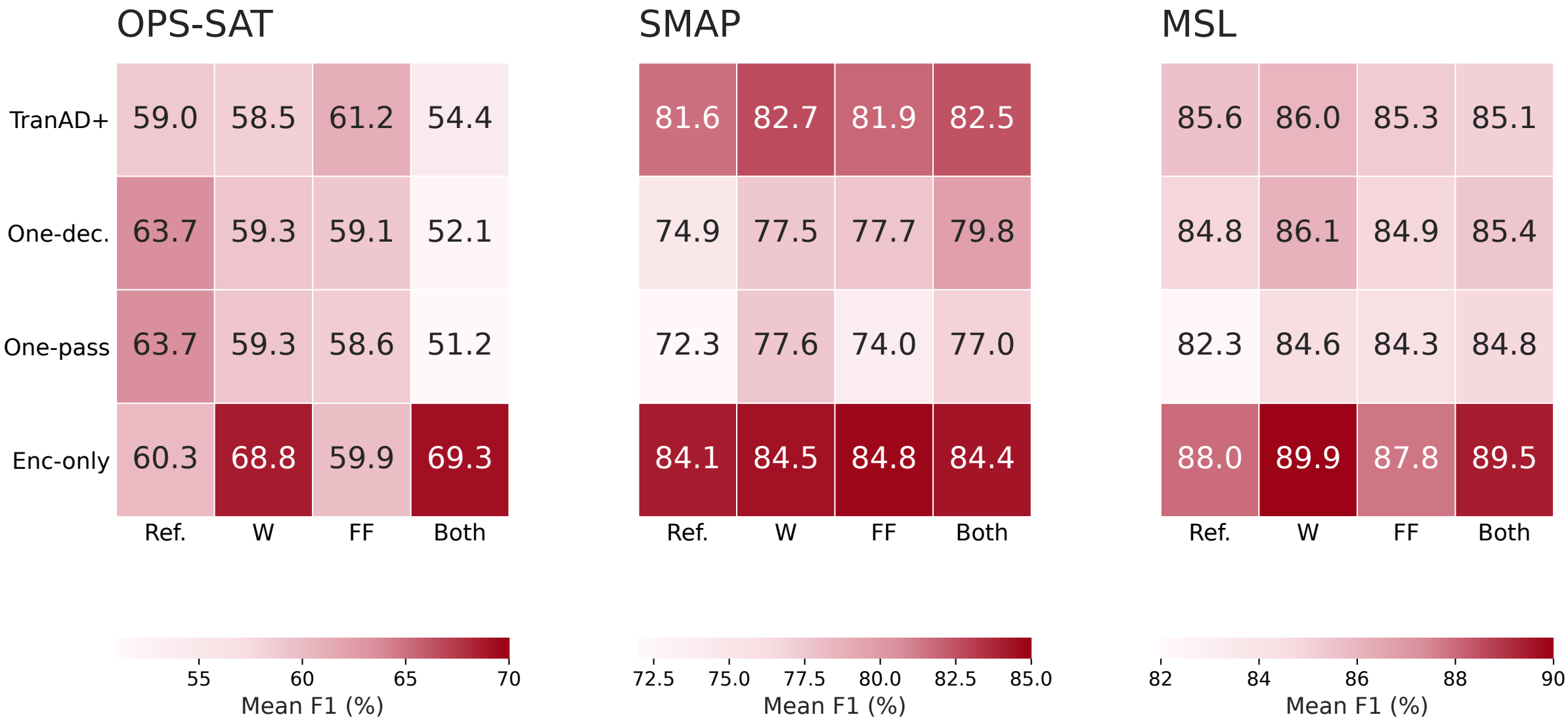
Reference W/FF · analytical matrix-operation counts; biases, activations and softmax excluded

Architectures × W/FF · BF200



FP32 · PA · 5 seeds per cell · dataset-specific colour scales

Architectures × W/FF · POT



FP32 · PA · 5 seeds per cell · dataset-specific colour scales

Quantized weights · FP32 calculations

Dataset	Method	FP32	INT8	INT4
OPS-SAT	BF200	85.55	85.21	83.61
OPS-SAT	POT	59.88	60.03	66.18
SMAP	BF200	81.38	81.37	80.34
SMAP	POT	79.82	79.61	76.73
MSL	BF200	46.71	46.71	46.69
MSL	POT	85.91	85.98	86.24

Mean F1 (%) · 4 architectures × 4 W/FF settings × 5 seeds per dataset · PA

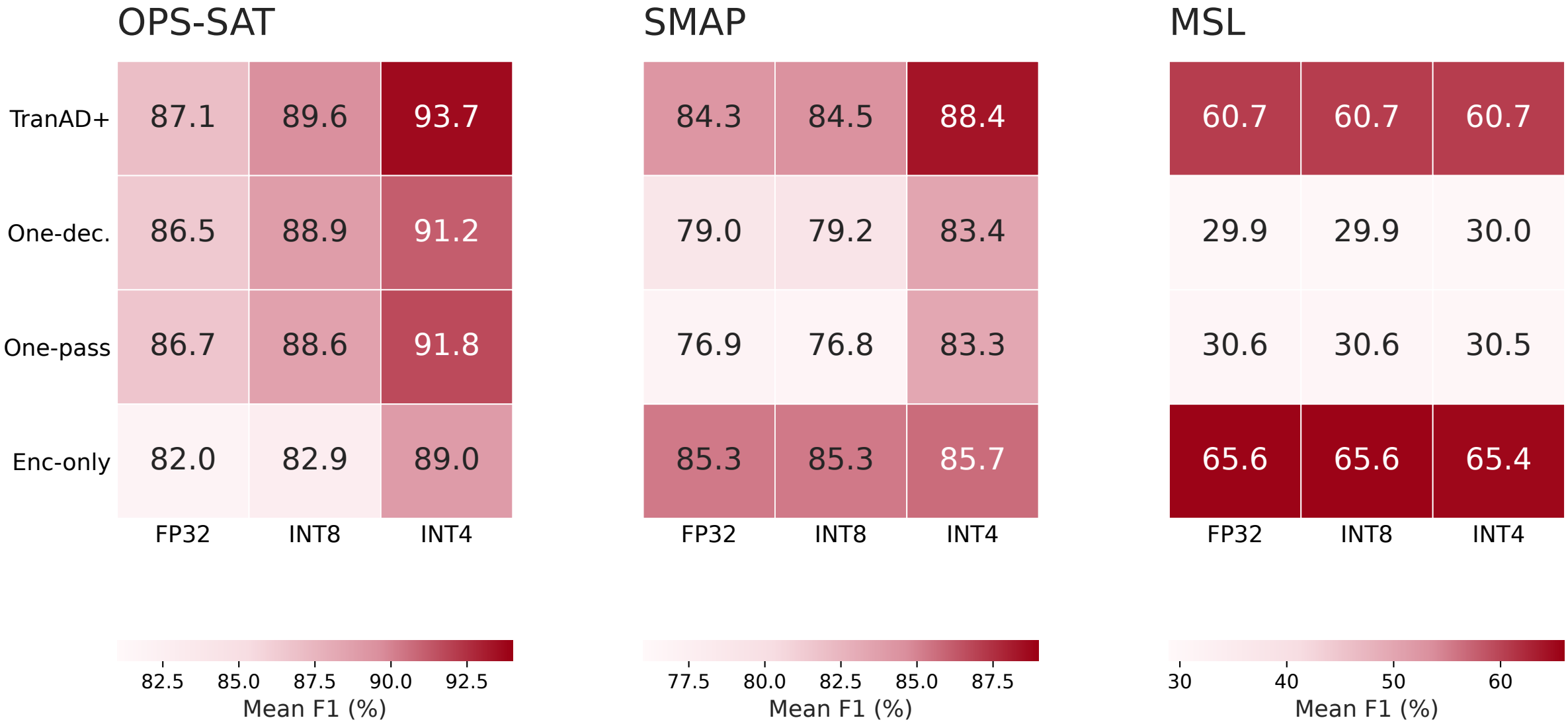
Quantized weights + integer matrix products

Dataset	Method	FP32	INT8	INT4
OPS-SAT	BF200	85.55	87.52	91.40
OPS-SAT	POT	59.88	62.23	83.09
SMAP	BF200	81.38	81.47	85.22
SMAP	POT	79.82	79.77	82.45
MSL	BF200	46.71	46.70	46.64
MSL	POT	85.91	85.88	89.22

INT8 = W8A8 · INT4 = W4A4 · W4A4 matrix products use INT8 instructions.

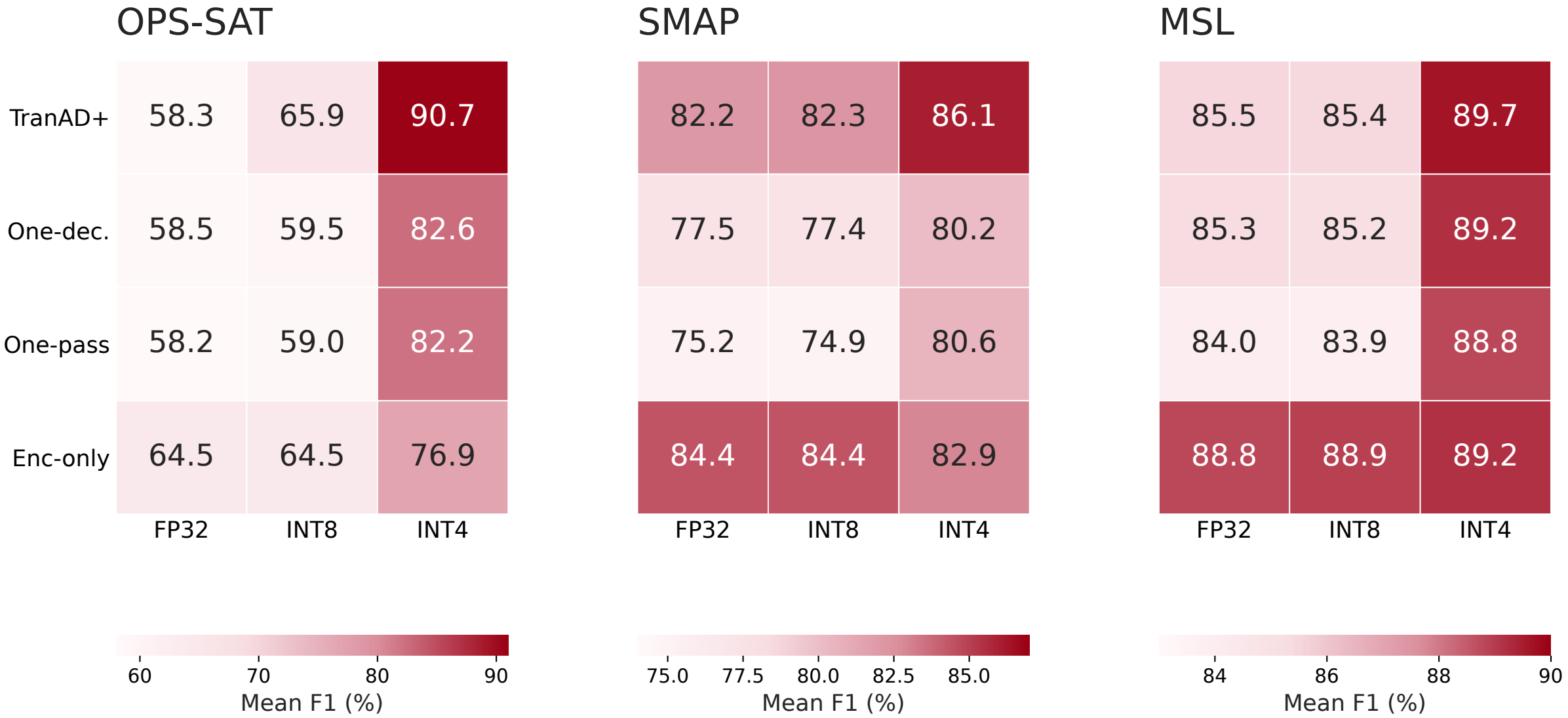
Mean F1 (%) · 4 architectures × 4 W/FF settings × 5 seeds per dataset · PA

Integer precision × architecture · BF200



PA · means over 4 W/FF settings × 5 seeds per cell · dataset-specific colour scales

Integer precision × architecture · POT



PA · means over 4 W/FF settings × 5 seeds per cell · dataset-specific colour scales

Integer quantization · F1 without PA

Dataset	Method	FP32	INT8	INT4
OPS-SAT	BF200	54.20	54.14	54.02
OPS-SAT	POT	4.72	4.84	4.65
SMAP	BF200	22.89	22.89	22.99
SMAP	POT	4.48	4.53	3.61
MSL	BF200	18.84	18.84	18.84
MSL	POT	16.09	16.21	12.23

Mean F1 (%) · 4 architectures × 4 W/FF settings × 5 seeds per dataset · raw-F1 selection

Integer quantization · stored model tensors (KiB)

Dataset	Model	FP32	INT8	INT4
OPS-SAT	TranAD+	1.1	0.8	0.7
	One-decoder	0.7	0.5	0.5
	One-pass	0.7	0.5	0.5
	Encoder-only	0.3	0.3	0.3
SMAP	TranAD+	265.9	77.0	44.7
	One-decoder	166.3	48.7	28.5
	One-pass	166.3	48.7	28.5
	Encoder-only	66.7	20.4	12.4
MSL	TranAD+	1285.4	357.4	200.8
	One-decoder	808.5	230.6	133.1
	One-pass	808.5	230.6	133.1
	Encoder-only	331.5	103.9	65.5

Reference W/FF · packed tensors, biases, scales and buffers included · training checkpoints excluded

Integer quantization · BF200 metrics (%)

Dataset	Format	Precision	Recall	F1	AUC-PA
OPS-SAT	FP32	82.41	89.99	85.55	88.84
	INT8	85.85	90.21	87.52	90.28
	INT4	91.58	91.54	91.40	93.19
SMAP	FP32	85.78	78.41	81.38	88.18
	INT8	85.35	78.89	81.47	88.38
	INT4	84.01	87.29	85.22	92.37
MSL	FP32	90.90	34.40	46.71	67.03
	INT8	90.93	34.40	46.70	67.03
	INT4	90.85	34.31	46.64	66.99

PA · mean over 4 architectures × 4 W/FF settings × 5 seeds · AUC-PA from binary predictions

Integer quantization · POT metrics (%)

Dataset	Format	Precision	Recall	F1	AUC-PA
OPS-SAT	FP32	92.44	44.62	59.88	71.27
	INT8	92.25	47.66	62.23	72.69
	INT4	94.63	75.35	83.09	86.50
SMAP	FP32	87.69	73.49	79.82	85.99
	INT8	87.58	73.49	79.77	85.98
	INT4	87.36	78.36	82.45	88.35
MSL	FP32	79.82	93.06	85.91	95.16
	INT8	79.81	93.02	85.88	95.14
	INT4	84.27	94.86	89.22	96.40

PA · mean over 4 architectures × 4 W/FF settings × 5 seeds · AUC-PA from binary predictions